Cartan spatial decay, rational trace, and variational-core reduction

TECT verification-first repository
claim `A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION`

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1 Purpose and scope

R-085 left two explicit analytic inputs: the complete mixed Cartan atom estimates (4.10)–(4.11), and the coupled rational shifted-Hessian form bound (6.5). R-086 then reduced the latter to one coefficient-dominant translated-Wick packet. This note advances both inputs and also removes an unnecessary downstream CORE burden.

The first result closes the *spatial* half of the Cartan atom. A fixed compactly supported smooth Littlewood–Paley decomposition gives exact principal support below Q_{j+4}. The complete remainder has conditional Hilbert–Schmidt decay

$$2^{-2s(m-k)}, \qquad \frac{1}{3} < \alpha < \frac{1}{2}, \quad \frac{1}{2} < s < 3\alpha - \frac{1}{2}. \tag{1.1}$$

At the production value $\alpha = 2/5$, the choice $s = 7/12$ has strict squared root and gap margins 7/30 and 13/30. This proves R-085 (4.10) for an explicit translated-model sequence q_k^{mod}. It does *not* prove the one-use summability (4.11). A rare-event fixture shows why pulling the translated $\mathcal{C}^\alpha$ norm outside expectation cannot prove it.

The second result is an exact η-regularised Schur/Wick completion of the R-086 rational packet. It removes the endpoint kernel for fixed $\eta > 0$, but creates a nonnegative trace debt of size at most η^{-1} times a quadratic Wick trace. The leading coefficient-dominant resonance remains.

The third result proves, at every fixed cutoff, that the Boue–Dupuis variational infimum is unchanged when restricted to bounded smooth cylindrical simple progressive controls. Therefore a uniform OVERLAP lower bound on that class would directly imply the $q = 10/9$ Nelson moment. A separate packetwise graph theorem for every finite-energy control is not needed for this scalar application. OVERLAP itself remains open.

The binding internal dependencies are R-075, R-081, R-084, R-085, and R-086. The deterministic paralinearisation is pinned to Bailleul–Bernicot and is reproduced below for the compactly supported Fourier convention used here. The variational formula is pinned to Hariya–Watanabe and Boue–Dupuis. A targeted audit of the external Contents archive found no theorem that closes the remaining adapted one-use or coefficient-dominant packet.

2 The complete mixed Cartan input atom

Let $Q_j = \{\xi \in \mathbb{Z}^3 : |\xi|_\infty \le 2^j\}$ be the physical sharp cutoff. For a production Cartan generator A, put

$$\Phi_{A,j} = P_{\Sigma_{j+1:J}} F_A, \qquad F_A(z) = q_A(z)z, \qquad q_A(z) = \frac{z^T S_A z}{|z|^2 + \epsilon_\rho}. \tag{2.1}$$

Target heat convolution is a probability convolution and therefore contracts derivative sup norms. The positive floor gives

$$\sup_{j,\Sigma} \max_{1\le r\le 5} \|D^r\Phi_{A,j}\|_\infty \le M_{\epsilon_\rho,5} < \infty. \tag{2.2}$$

Condition on $\mathcal{F}_{j-1}$. Write

$$w_j = U_{j-1} + g_j, \qquad u_{j,r} = w_j + rA^{(j)}, \qquad A^{(j)} = A^{(\ell)} + \sum_{k=j_0}^{j} a_k, \qquad a_k = K_k h_k. \tag{2.3}$$

All fields in (2.3) are supported in Q_j. Linearity of the a and Da slots in R-085 (3.1) gives the exact atomisation

$$\mathbb{D}_j H_{A,j,A^{(j)}} = \mathcal{M}_{A,j,\ell} + \sum_{k=j_0}^{j} \mathcal{M}_{A,j,k}, \tag{2.4}$$

where the finite-low atom $\mathcal{M}_{A,j,\ell}$ stays under the accepted R-080 low-object bound, and

$$\begin{aligned}\mathcal{M}_{A,j,k}[v] = \int_0^1 \{ & D^2\Phi_{A,j}(u_{j,r})[a_k, v]^T Du_{j,r} \\ & + D\Phi_{A,j}(u_{j,r})[v]^T Da_k + D\Phi_{A,j}(u_{j,r})[a_k]^T Dv\}\, dr.\end{aligned} \tag{2.5}$$

No channel in (2.5) is discarded. Predictability says $\mathbb{D}_j a_k = 0$; it does not say $Da_k = 0$.

3 A reproduced order-two paralinearisation lemma

Fix $\chi \in C_c^\infty(\mathbb{R}^3)$, equal to one on $\{|\xi|_\infty \le 3/4\}$ and supported in $\{|\xi|_\infty \le 4/3\}$. Put

$$S_n = \chi(2^{-n}D), \qquad \Delta_n = S_n - S_{n-1}, \qquad T_f g = \sum_n S_{n-2} f\, \Delta_n g. \tag{3.1}$$

These smooth analytic blocks are distinct from, but compatible with, the sharp physical filtration Q_j. They are used only to decompose the exact finite-cutoff function.

For a smooth vector- or tensor-valued map G, define

$$\mathcal{P}_1[G,u] = T_{DG(u)} u, \tag{3.2}$$

and

$$\mathcal{P}_2[G,u] = T_{DG(u)} u + \frac{1}{2}\{T_{D^2G(u)}(u\otimes u) - 2T_{D^2G(u)[u]} u\}. \tag{3.3}$$

Lemma 3.1 (order-one and order-two remainders)

If $0<\alpha<1$, $u \in \mathcal{C}^\alpha$, and the derivatives of G needed below are bounded, then, componentwise,

$$\|\Delta_n\{G(u) - \mathcal{P}_1[G,u]\}\|_\infty \le C_{G,\alpha} 2^{-2\alpha n}(1+\|u\|_{\mathcal{C}^\alpha})^2, \tag{3.4}$$

$$\|\Delta_n\{G(u) - \mathcal{P}_2[G,u]\}\|_\infty \le C_{G,\alpha} 2^{-3\alpha n}(1+\|u\|_{\mathcal{C}^\alpha})^3. \tag{3.5}$$

Proof. Let K_n be the kernel of Δ_n. Its zero integral lets one write each block of $G(u)$ as an integral of $G(u(y)) - G(u(x))$. Insert the first-order Taylor formula around the low-frequency value $S_{n-2}u(x)$. Subtracting $S_{n-2}DG(u)\Delta_n u$ leaves two increments of u; each is bounded by $C2^{-\alpha n}\|u\|_{\mathcal{C}^\alpha}$. The uniformly integrable smooth kernels give (3.4).

For the next order, expand once more. The quadratic Taylor contribution is exactly

$$\frac{1}{2}\{T_{D^2G(u)}(u\otimes u)-2T_{D^2G(u)[u]}u\}.$$

The two terms are both required: they remove the dependence on the chosen low-frequency base point. The integral remainder contains three increments of u, hence $2^{-3\alpha n}$. Summing the finitely overlapping smooth kernels proves (3.5). This is the compact Fourier version of the high-order Taylor expansion in Bailleul–Bernicot, Forum of Mathematics, Sigma 7 (2019), e44, Section 2 and Appendix B. ☐

Apply (3.4) to $G=D^2\Phi$ and (3.5) to $G=D\Phi$. With

$$\mathcal{R}_2(u)=D^2\Phi(u)-\mathcal{P}_1[D^2\Phi,u],\qquad \mathcal{R}_3(u)=D\Phi(u)-\mathcal{P}_2[D\Phi,u], \tag{3.6}$$

and $Z=1+\|u\|_{\mathcal{C}^\alpha}$, equations (2.2), (3.4), and (3.5) give

$$\|\Delta_n\mathcal{R}_2(u)\|_\infty\le C_e2^{-2\alpha n}Z^2,\qquad \|\Delta_n\mathcal{R}_3(u)\|_\infty\le C_e2^{-3\alpha n}Z^3. \tag{3.7}$$

4 Exact principal support and the safe separation

The explicit mask in (3.1) fixes the support constant, rather than hiding it in $O(1)$. A Q_j-supported input has no Δ_n block above $j+1$; its square has no block above $j+2$. The support of one paraproduct summand is contained in $(5/3)Q_n$. Therefore

$$\operatorname{supp}\mathcal{P}_1[D^2\Phi,u]\subset\frac{10}{3}Q_j,\qquad \operatorname{supp}\mathcal{P}_2[D\Phi,u]\subset\frac{20}{3}Q_j. \tag{4.1}$$

After inserting the remaining factors in (2.5), the first principal channel has radius at most $(10/3+3)2^j=19\,2^j/3$. The other two have radius at most $(20/3+2)2^j=26\,2^j/3$. Hence every principal contribution lies in Q_{j+4}. Since Π_m is supported where $|\xi|_\infty>2^{m-1}$,

$$\boxed{\Pi_m\mathcal{M}^{\text{principal}}_{A,j,k}=0\quad(m\ge j+5).} \tag{4.2}$$

Thus $C_0=5$ is a safe explicit constant for the declared smooth analytic partition. The sharper value four belongs to a formal sharp paraproduct and is not used, because sharp partial-sum kernels are not uniformly L^1. Conditional OU averaging and target heat act in probability and target variables, respectively, so both preserve (4.2).

There is no algebraic cancellation claim here. On the two-dimensional slice $\Phi(z)=q(z)z$, $q=(x^2-y^2)/(x^2+y^2+e)$, the coefficient one-form has

$$d(\Phi\cdot dz)=\frac{4xy}{x^2+y^2+e}\,dx\wedge dy. \tag{4.3}$$

For $w=(1,1)$, $a=\varepsilon\sin(Kx)e_1$, and $v=\cos(Jx)e_2$, the first-order mixed principal is

$$-\frac{\varepsilon}{2+e}\{(K+J)\cos((J-K)x)+(K-J)\cos((J+K)x)\}. \tag{4.4}$$

It is nonzero, but has only the sum and difference carriers and is therefore killed in the far region (4.2).

5 Hilbert–Schmidt decay of the complete remainder

Let $(v_h)_h$ be an orthonormal fresh-root Cameron–Martin basis. The three-dimensional $|q|^{-4}$ shell covariance gives the exact scale powers

$$\sup_x \sum_h |v_h(x)|^2 \le \kappa_0\, 2^{-j}, \qquad \sup_x \sum_h |Dv_h(x)|^2 \le \kappa_1\, 2^{j}. \tag{5.1}$$

Band limitation also gives

$$\|Du_{j,r}\|_\infty \le C_\alpha 2^{(1-\alpha)j} Z_{j,r}, \qquad Z_{j,r} = 1 + \|u_{j,r}\|_{\mathcal{C}^\alpha}. \tag{5.2}$$

When $m \ge j+5$, Fourier support forces the output frequency into $\mathcal{R}_2$ or $\mathcal{R}_3$. Square (3.7), sum the fresh roots pointwise using (5.1), and then integrate the control factor in L^2. This gives

$$\begin{aligned}\sum_h \|\Pi_m[\mathcal{R}_2(u)[a_k, v_h]^T Du]\|_2^2 &\le C_e 2^{-4\alpha m} 2^{(1-2\alpha)j} Z^6 \|a_k\|_2^2,\\ \sum_h \|\Pi_m[\mathcal{R}_3(u)[v_h]^T Da_k]\|_2^2 &\le C_e 2^{-6\alpha m} 2^{-j} Z^6 \|Da_k\|_2^2,\\ \sum_h \|\Pi_m[\mathcal{R}_3(u)[a_k]^T Dv_h]\|_2^2 &\le C_e 2^{-6\alpha m} 2^{j} Z^6 \|a_k\|_2^2.\end{aligned} \tag{5.3–5.5}$$

There is no 2^{3k} Bernstein loss in (5.3): after the pointwise root sum, a_k remains in L^2. Cross-channel orthogonality is not assumed; the elementary three-term square inequality pays a fixed factor.

Put $\beta = 6\alpha - 1$ and $d = m - j$. The two critical right-hand sides become

$$C_e 2^{-\beta j} 2^{-4\alpha d} Z^6 \|a_k\|_2^2, \qquad C_e 2^{-\beta j} 2^{-6\alpha d} Z^6 \|a_k\|_2^2. \tag{5.6}$$

The middle channel is stronger. Conditional OU contraction gives

$$\int_0^\infty e^{-2\tau} \mathbb{E}\|P_\tau^{(j)} X\|^2 d\tau \le \frac{1}{2}\mathbb{E}\|X\|^2. \tag{5.7}$$

Jensen in r and (5.3)–(5.7) now prove the spatial atom theorem.

Theorem 5.1 (the spatial half of R-085 (4.10))

Assume $1/3 < \alpha < 1/2$ and $1/2 < s < 3\alpha - 1/2$. The upper restriction on α implies $3\alpha - 1/2 < 2\alpha$, so both decay conditions $\beta - 2s > 0$ and $4\alpha - 2s > 0$ hold. Define

$$W_k = \sup_{j \ge k} \sum_A \mathbb{E} \int_0^1 Z_{j,r}^6 (\|a_k\|_2^2 + \|Da_k\|_2^2)\, dr, \tag{5.8}$$

$$q_k^{\mathrm{mod}} = C_e 2^{-(6\alpha-1)k} W_k. \tag{5.9}$$

Then for $m \ge j+5$,

$$\boxed{\sum_A \int_0^\infty e^{-2\tau} \mathbb{E}\|\Pi_m P_\tau^{(j)} \mathcal{M}_{A,j,k}\|_{\mathrm{HS}}^2 d\tau \le 2^{-2s(m-k)} q_k^{\mathrm{mod}}.} \tag{5.10}$$

Indeed, after writing $j = k + r$, the excess powers are

$$(\beta - 2s)r + (4\alpha - 2s)d \ge 0. \tag{5.11}$$

At $\alpha = 2/5$ and $s = 7/12$,

$$\beta = \frac{7}{5}, \qquad \beta - 2s = \frac{7}{30}, \qquad 4\alpha - 2s = \frac{13}{30}, \tag{5.12}$$

so the R-085 nonorthogonal Schur threshold is crossed with strict room.

6 The exact Cartan one-use obstruction

The Cameron–Martin multiplier gives

$$\|a_k\|_2^2 \lesssim 2^{-4k}\|h_k\|_2^2, \qquad \|Da_k\|_2^2 \lesssim 2^{-2k}\|h_k\|_2^2. \tag{6.1}$$

Using (6.1) in (5.9) and extracting the translated model norm would require

$$\sum_k 2^{-(\beta+1)k} \sup_{j\geq k} \mathbb{E}[Z_j^6\|h_k\|_2^2] \lesssim 1 + \mathbb{E}[X^{1/2}Y^{1/2}]. \tag{6.2}$$

This implication is false as a general energy argument. Let an event E have probability $p = N^{-6}$ and take one predictable physical mode

$$h = N^3\mathbf{1}_E e_N, \qquad a = K_N h = N\mathbf{1}_E e_N. \tag{6.3}$$

Then $\mathbb{E}X \asymp \mathbb{E}Y \asymp \mathbb{E}\sqrt{XY} \asymp 1$, while $Z \asymp N^{1+\alpha}$ on E. Since $\beta+1 = 6\alpha$, the corresponding extracted summand in (6.2) is of order $N^6 = p^{-1}$.

This registers

$$\texttt{NG-2026-07-25-A13-PATHWISE-TRANSLATED-MODEL-NORM-EXTRACTION.} \tag{6.4}$$

It is a proof-method no-go, not a counterexample to (5.10) or to the directly averaged complete atom. The remaining Cartan lemma must keep expectation inside the three exact remainders and prove

$$\sum_k 2^k q_k \lesssim 1 + \mathbb{E}[X^{1/2}Y^{1/2}] \tag{6.5}$$

without extracting an unbounded translated model multiplier. Thus R-085 (4.11), controlled Cartan CFAR, and REG remain open.

7 Exact rational eta-completion and trace debt

Use the R-086 notation

$$\mathcal{P} = \frac{1}{2}L : Q + G^T Lc + \frac{1}{2}c^T B_1 c, \qquad Q = G\otimes G - \Gamma. \tag{7.1}$$

For $\eta > 0$, put

$$A_\eta = B_1 + 2\eta I, \qquad M_\eta = L - LA_\eta^{-1}L. \tag{7.2}$$

Since $B_1 \succeq 0$, $A_\eta \succ 0$ even on the endpoint kernel. Completing the c square gives the exact identity

$$\boxed{\begin{aligned}\mathcal{P} + \eta|c|^2 = {}& \frac{1}{2}(c + A_\eta^{-1}LG)^T A_\eta (c + A_\eta^{-1}LG) \\ & + \frac{1}{2}M_\eta : Q - \frac{1}{2}\operatorname{Tr}(LA_\eta^{-1}L\Gamma).\end{aligned}} \tag{7.3}$$

Expanding the square proves (7.3); the two Γ contractions recombine to $-L : \Gamma/2$. For $\Gamma \succeq 0$ the trace debt

$$\mathfrak{D}_\eta = \frac{1}{2}\operatorname{Tr}(LA_\eta^{-1}L\Gamma) \tag{7.4}$$

is nonnegative, and congruence of $A_\eta^{-1} \preceq (2\eta)^{-1}I$ gives

$$0 \le \mathfrak{D}_\eta \le \frac{1}{4\eta}\operatorname{Tr}(L^2\Gamma). \tag{7.5}$$

This is a useful kernel-free normal form but not a form bound. In general M_η is indefinite. If $L = O(a^3)$ at fixed η, then

$$M_\eta = L + O_\eta(a^6), \tag{7.6}$$

so the leading coefficient-dominant $L : Q$ resonance survives. The old $O(\sigma^{-2})$ inverse-heat loss has been traded for an $O(\eta^{-1})$ trace debt. No cutoff-uniform bound on that debt or on the complete rational packet is asserted.

8 Fixed-cutoff bounded cylindrical variational core

Fix a cutoff J. Let B be a finite-dimensional Brownian motion on its completed canonical filtration, let H_J be finite-dimensional, and let

$$X_J = \int_0^T K_J(t)\,dB_t, \qquad T_J v = \int_0^T K_J(t) v_t\,dt, \qquad K_J \in L_t^2. \tag{8.1}$$

Assume $G_J : H_J \to \mathbb{R}$ is continuous, bounded below, and, for every $R < \infty$, obeys the local polynomial envelope

$$\sup_{\|y\|\le R} |G_J(x+y)| \le C_{J,R}(1 + \|x\|^{r_J}). \tag{8.2}$$

Let $\mathcal{A}$ be the finite-energy progressive controls, $\mathcal{A}_b$ the bounded progressive controls, and $\mathcal{A}_{cs}$ the bounded smooth cylindrical simple progressive controls. Thus elements of $\mathcal{A}_{cs}$ have the form

$$v_t = \sum_{\ell=1}^N \Psi_\ell(B_{r_{\ell,1}}, \dots, B_{r_{\ell,m_\ell}})\mathbf{1}_{(t_{\ell-1},t_\ell]}(t), \qquad r_{\ell,i} \le t_{\ell-1}, \tag{8.3}$$

with bounded smooth Ψ_ℓ.

Theorem 8.1 (fixed-cutoff variational CORE)

For every $q > 0$,

$$\boxed{-\frac{1}{q}\log \mathbb{E}e^{-qG_J(X_J)} = \inf_{v\in\mathcal{A}} \mathbb{E}\left[G_J(X_J + T_J v) + \frac{1}{2q}\int_0^T |v_t|^2 dt\right]} \tag{8.4}$$

and the infimum is unchanged when $\mathcal{A}$ is replaced by $\mathcal{A}_b$ or $\mathcal{A}_{cs}$.

Proof. Put $F = -qG_J(X_J)$. If $m_J = \inf G_J$, then $F \le -qm_J$ and e^F is integrable. The polynomial envelope and Gaussian moments give $\mathbb{E}F^- < \infty$. Hariya–Watanabe, Theorem 1.1 and Corollary 2.1, therefore give the Boue–Dupuis identity and permit restriction to bounded drifts. This uses only the fixed-J lower bound and local polynomial envelope; no Nelson estimate is assumed.

For $v \in \mathcal{A}_b$, bounded simple progressive approximants v_n can be chosen with the same deterministic bound and $v_n \to v$ in $L^2(\Omega \times [0,T])$. Hence

$$\|T_J(v_n - v)\|_{L^2(\Omega;H_J)} \le \|K_J\|_{L_t^2}\|v_n - v\|_{L^2_{\Omega,t}}, \tag{8.5}$$

and the quadratic costs converge. Also $\|T_J v_n\| \le M\int_0^T \|K_J(t)\|_{\mathrm{op}}dt$. Continuity, convergence in probability, and the common Gaussian-polynomial dominator from (8.2) imply $G_J(X_J + T_J v_n) \to G_J(X_J + T_J v)$ in L^1.

Finally approximate each bounded past-measurable simple coefficient in L^2 by bounded smooth functions of finitely many past Brownian evaluations. Repeating (8.5) proves equality with the cylindrical-simple infimum. Independent auxiliary randomisation cannot lower the value: condition on the complete independent seed, apply the canonical lower bound sectionwise, and integrate. Correlated future information or auxiliary randomness in the payoff is not covered. □

At the A13 values

$$q=\frac{10}{9},\qquad \frac{1}{2q}=\frac{9}{20},\qquad q-\frac{11}{10}=\frac{1}{90},\qquad \frac{1}{2(11/10)}-\frac{9}{20}=\frac{1}{220}. \tag{8.6}$$

Thus a cutoff-, partition-, and revisit-uniform OVERLAP estimate $I_{J,10/9}(v)\ge -C$ for $v\in\mathcal{A}_{cs}$ would imply

$$\mathbb{E}e^{-(10/9)G_J}\le e^{(10/9)C}. \tag{8.7}$$

The theorem bypasses a direct all-control *packetwise* extension. It does not prove OVERLAP, termwise packet convergence, graph recovery for every finite-energy control, or the Nelson lower bound itself.

9 Proof-and-failure map

Branch	**Verdict**	**Next consequence**
Complete Cartan atomisation	Exact	(2.4)–(2.5) retain all channels
Order-two principal	Far-supported	(4.2), safe $C_0=5$
Cartan remainder spatial decay	Proved	(5.10), every $1/2<s<7/10$
Translated model-norm extraction	Failed method	(6.3)–(6.4), growth p^{-1}
Cartan one-use ledger	Open	Prove expectation-inside (6.5)
Rational endpoint kernel	Regularised	(7.3), fixed $\eta>0$
Rational trace debt	Open	Control (7.4) with complete Wick forest
Fixed-cutoff variational CORE	Proved	Invoke after uniform OVERLAP
REG and OVERLAP	Open	Reassemble after both analytic packets close

The revised dependency line is

$$\text{Cartan (4.11)} + \text{rational packet} \Longrightarrow \text{REG} \Longrightarrow \text{OVERLAP} \Longrightarrow \text{invoke Theorem 8.1} \Longrightarrow \text{Nelson.} \tag{9.2}$$

10 Devil's-advocate review

1. **"The sharp physical cutoff makes Lemma 3.1 automatic with $C_0=4$." VALID WITH MITIGATION.** Sharp partial-sum kernels are not uniformly L^1. The proof instead declares a smooth analytic LP partition and uses the conservative, verified value $C_0=5$.

2. **"The three Cartan channels cancel algebraically." DISMISSED.** The curvature and two-mode fixture (4.3)–(4.4) are nonzero. The accepted mechanism is far-support annihilation plus remainder decay.

3. **"Equation (5.10) proves controlled Cartan CFAR." UPHELD as false.** R-085 also requires the one-use sum (6.5). That ledger is explicitly open.

4. **“The rare-event fixture disproves the production atom.” DISMISSED.** It tests only the stronger pathwise extraction (6.2). The expectation-inside Fourier/Wick estimate may still hold.

5. **“Adding $\eta|c|^2$ closes rational NEAR.” UPHELD as false.** The trace debt is $O(\eta^{-1})$, M_η is indefinite, and the leading $L : Q$ packet remains.

6. **“Density plus lower semicontinuity transfers every packet lower bound to all controls.” UPHELD as false.** Lower semicontinuity gives the wrong inequality direction for that claim. Theorem 8.1 instead identifies the scalar variational infimum using the bounded-drift version of Boue–Dupuis and L^1 payoff convergence.

7. **“CORE proves Nelson now.” UPHELD as false.** The complete uniform OVERLAP lower bound remains missing.

11 Authorities and executable verification

External analytic authorities:

- I. Bailleul and F. Bernicot, “High order paracontrolled calculus,” Forum Math. Sigma 7 (2019), e44, arXiv:1609.06966, doi:10.1017/fms.2019.44.
- Y. Hariya and S. Watanabe, “The Boue–Dupuis formula and the exponential hypercontractivity in the Gaussian space,” Electron. Commun. Probab. 27 (2022), Paper 13, arXiv:2110.14852, doi:10.1214/22-ECP461.
- M. Boue and P. Dupuis, “A variational representation for certain functionals of Brownian motion,” Ann. Probab. 26 (1998), 1641–1659, doi:10.1214/aop/1022855876.

The primary executable checks the support radii, every exact exponent margin, Cartan curvature, rare-event scaling, symbolic rational completion, trace debt bounds, small-amplitude order, and A13 variational coefficients. The non-importing audit uses handwritten rational matrix algebra, a noncommuting three-dimensional fixture, mutation tests, seeded PSD stress tests, a finite-difference/FFT two-mode Cartan check, rare events, and a solvable linear Gaussian variational fixture.

Run from the repository root:

```
python codes/foundations/
  a13_classii_cartan_spatial_decay_rational_trace_
  variational_core_reduction_verify.py
```

The integrated verifier pins authorities and evidence hashes, reruns both children, inspects the proof PDF and tracked status surfaces, and fails closed on every open downstream flag.

Result footer

```
Result ID: A13-CLASSII-CARTAN-SPATIAL-DECAY-RATIONAL-TRACE-
           VARIATIONAL-CORE-REDUCTION
Precise statement:        The complete mixed Cartan atom has exact far
                          principal support and remainder decay for every
                          1/3<alpha<1/2 and 1/2<s<3alpha-1/2; at alpha=2/5,
                          s=7/12 has margins
                          7/30 and 13/30. This proves the spatial half (4.10)
                          with an explicit q_k^mod, not the one-use ledger
                          (4.11). The rational packet has an exact eta-Schur/
                          Wick completion with a retained trace debt. At each
```

	fixed cutoff, the Boue--Dupuis infimum equals its bounded smooth cylindrical-simple restriction.
Scope:	Finite cutoff; fixed positive floor; declared smooth analytic LP partition; canonical Brownian filtration; continuous bounded-below payoff with the local polynomial envelope of Theorem 8.1.
Dependencies:	R-075; R-081; R-084--R-086; cited external theorems.
Evidence grade:	ANALYTIC, EXACT, EXECUTED.
Reproduction command:	python codes/foundations/a13_classii_cartan_spatial_decay_rational_trace_variational_core_reduction_verify.py
Expected output:	Primary, non-importing independent, integrated, and aggregate manifest-pinned PASS; exit zero.
Negative result:	NG-2026-07-25-A13-PATHWISE-TRANSLATED-MODEL-NORM-EXTRACTION (method no-go only).
Falsification gate:	Failure of a paralinearisation remainder, support radius, HS power, exponent, rational identity, CORE hypothesis, evidence hash, PDF, or assertion contract.
Tier before / after:	T4 / T4; no promotion.
No-overclaim statement:	No Cartan one-use q-ledger, complete production atom, controlled Cartan CFAR, coefficient-dominant rational bound, rational or signed NEAR, REG, OVERLAP, Nelson, measure theorem, removal limit, or Sector-A closure.
Next required action:	Prove the expectation-inside Cartan one-use lemma and the rational coefficient-dominant trace/model packet; then reassemble REG and prove uniform OVERLAP.